

# False Position (Regula Falsi) Method

Solved Questions — Correct to Four Decimal Places

## Q.5 (i) Find a root of $x^3 - 4x - 9 = 0$ using False Position Method.

Given,  $f(x) = x^3 - 4x - 9$

Take  $a = 2$ ,  $b = 3$

$$f(2) = 8 - 8 - 9 = -9$$

$$f(3) = 27 - 12 - 9 = 6$$

Since  $f(2)f(3) < 0$ , a root lies between 2 and 3.

**False Position Formula:**  $x = [a f(b) - b f(a)] / [f(b) - f(a)]$

$$\text{Iteration 1: } x_1 = [2(6) - 3(-9)] / [6 - (-9)] = 2.6000$$

Since  $f(2.6) < 0$ , take  $a = 2.6$ ,  $b = 3$ .

| Iteration | a      | b      | x      |
|-----------|--------|--------|--------|
| 1         | 2.0000 | 3.0000 | 2.6000 |
| 2         | 2.6000 | 3.0000 | 2.6933 |
| 3         | 2.6933 | 3.0000 | 2.7049 |
| 4         | 2.7049 | 3.0000 | 2.7063 |
| 5         | 2.7063 | 3.0000 | 2.7065 |

**Therefore,**  $x = 2.7065$  (correct to four decimal places).

## Q.5 (ii) Find a root of $x^3 + 2x^2 + 10x - 20 = 0$ using False Position Method.

Given,  $f(x) = x^3 + 2x^2 + 10x - 20$

Take  $a = 1$ ,  $b = 2$

$$f(1) = 1 + 2 + 10 - 20 = -7$$

$$f(2) = 8 + 8 + 20 - 20 = 16$$

Since  $f(1)f(2) < 0$ , a root lies between 1 and 2.

**False Position Formula:**  $x = [a f(b) - b f(a)] / [f(b) - f(a)]$

$$\text{Iteration 1: } x_1 = [1(16) - 2(-7)] / [16 - (-7)] = 1.3043$$

Since  $f(1.3043) < 0$ , take  $a = 1.3043$ ,  $b = 2$ .

| Iteration | a      | b      | x      |
|-----------|--------|--------|--------|
| 1         | 1.0000 | 2.0000 | 1.3043 |
| 2         | 1.3043 | 2.0000 | 1.3579 |
| 3         | 1.3579 | 2.0000 | 1.3670 |
| 4         | 1.3670 | 2.0000 | 1.3685 |
| 5         | 1.3685 | 2.0000 | 1.3688 |
| 6         | 1.3688 | 2.0000 | 1.3688 |

**Therefore,**  $x = 1.3688$  (correct to four decimal places).

**Q.5 (iii) Find a root of  $x^3 - 4x - 1 = 0$  using False Position Method.**

Given,  $f(x) = x^3 - 4x - 1$

Take  $a = 2$ ,  $b = 3$

$$f(2) = 8 - 8 - 1 = -1$$

$$f(3) = 27 - 12 - 1 = 14$$

Since  $f(2)f(3) < 0$ , a root lies between 2 and 3.

**False Position Formula:**  $x = [a f(b) - b f(a)] / [f(b) - f(a)]$

$$\text{Iteration 1: } x_1 = [2(14) - 3(-1)] / [14 - (-1)] = 2.0667$$

Since  $f(2.0667) < 0$ , take  $a = 2.0667$ ,  $b = 3$ .

| Iteration | a      | b      | x      |
|-----------|--------|--------|--------|
| 1         | 2.0000 | 3.0000 | 2.0667 |
| 2         | 2.0667 | 3.0000 | 2.0951 |
| 3         | 2.0951 | 3.0000 | 2.1068 |
| 4         | 2.1068 | 3.0000 | 2.1116 |
| 5         | 2.1116 | 3.0000 | 2.1136 |
| 6         | 2.1136 | 3.0000 | 2.1144 |
| 7         | 2.1144 | 3.0000 | 2.1147 |
| 8         | 2.1147 | 3.0000 | 2.1148 |
| 9         | 2.1148 | 3.0000 | 2.1149 |

**Therefore,**  $x = 2.1149$  (correct to four decimal places).

**Q.5 (iv) Find a root of  $x^6 - x^4 - x^3 - 1 = 0$  using False Position Method.**

Given,  $f(x) = x^6 - x^4 - x^3 - 1$

Take  $a = 1.4$ ,  $b = 1.41$

$$f(1.4) \approx -0.0561$$

$$f(1.41) \approx 0.1023$$

Since  $f(1.4)f(1.41) < 0$ , a root lies between 1.4 and 1.41.

**False Position Formula:**  $x = [a f(b) - b f(a)] / [f(b) - f(a)]$

$$\text{Iteration 1: } x_1 \approx [1.4(0.1023) - 1.41(-0.0561)] / [0.1023 - (-0.0561)] \approx 1.4035$$

| Iteration | a      | b      | x      |
|-----------|--------|--------|--------|
| 1         | 1.4000 | 1.4100 | 1.4035 |
| 2         | 1.4035 | 1.4100 | 1.4036 |
| 3         | 1.4036 | 1.4100 | 1.4036 |

**Therefore,**  $x = 1.4036$  (correct to four decimal places).

**Q.5 (v) Find a root of  $xe^x = 2$  using False Position Method.**

Given,  $xe^x - 2 = 0$

Let  $f(x) = xe^x - 2$

Take  $a = 0$ ,  $b = 1$

$f(0) = 0(e^0) - 2 = -2$

$f(1) = e - 2 \approx 0.7183$

Since  $f(0)f(1) < 0$ , a root lies between 0 and 1.

**False Position Formula:**  $x = [a f(b) - b f(a)] / [f(b) - f(a)]$

Iteration 1:  $x_1 = [0(0.7183) - 1(-2)] / [0.7183 - (-2)] \approx 0.7358$

| Iteration | a      | b      | x      |
|-----------|--------|--------|--------|
| 1         | 0.0000 | 1.0000 | 0.7358 |
| 2         | 0.7358 | 1.0000 | 0.8395 |
| 3         | 0.8395 | 1.0000 | 0.8512 |
| 4         | 0.8512 | 1.0000 | 0.8525 |
| 5         | 0.8525 | 1.0000 | 0.8526 |
| 6         | 0.8526 | 1.0000 | 0.8526 |

**Therefore,**  $x = 0.8526$  (correct to four decimal places).

**Q.5 (vi) Find a root of  $e^x \sin x = 1$  using False Position Method.**

Given,  $e^x \sin x - 1 = 0$

Let  $f(x) = e^x \sin x - 1$

Take  $a = 0$ ,  $b = 1$

$f(0) = e^0 \sin 0 - 1 = -1$

$f(1) = e \sin 1 - 1 \approx 1.2874$

Since  $f(0)f(1) < 0$ , a root lies between 0 and 1.

**False Position Formula:**  $x = [a f(b) - b f(a)] / [f(b) - f(a)]$

Iteration 1:  $x_1 = [0(1.2874) - 1(-1)] / [1.2874 - (-1)] \approx 0.4372$

| Iteration | a      | b      | x      |
|-----------|--------|--------|--------|
| 1         | 0.0000 | 1.0000 | 0.4372 |
| 2         | 0.4372 | 1.0000 | 0.5560 |
| 3         | 0.5560 | 1.0000 | 0.5819 |
| 4         | 0.5819 | 1.0000 | 0.5872 |
| 5         | 0.5872 | 1.0000 | 0.5883 |
| 6         | 0.5883 | 1.0000 | 0.5885 |
| 7         | 0.5885 | 1.0000 | 0.5885 |

**Therefore,**  $x = 0.5885$  (correct to four decimal places).

**Q.5 (vii) Find a root of  $x = \cos x$  using False Position Method.**

Given,  $x - \cos x = 0$

Let  $f(x) = x - \cos x$

Take  $a = 0$ ,  $b = 1$

$$f(0) = 0 - \cos 0 = -1$$

$$f(1) = 1 - \cos 1 = 1 - 0.5403 = 0.4597$$

Since  $f(0)f(1) < 0$ , a root lies between 0 and 1.

**False Position Formula:**  $x = [a f(b) - b f(a)] / [f(b) - f(a)]$

$$\text{Iteration 1: } x_1 = [0(0.4597) - 1(-1)] / [0.4597 - (-1)] \approx 0.6851$$

| Iteration | a      | b      | x      |
|-----------|--------|--------|--------|
| 1         | 0.0000 | 1.0000 | 0.6851 |
| 2         | 0.6851 | 1.0000 | 0.7363 |
| 3         | 0.7363 | 1.0000 | 0.7389 |
| 4         | 0.7389 | 1.0000 | 0.7391 |
| 5         | 0.7391 | 1.0000 | 0.7391 |

**Therefore,**  $x = 0.7391$  (correct to four decimal places).

**Q.5 (viii) Find a root of  $x \tan x = -1$  in (2.5, 3) using False Position Method.**

Given,  $x \tan x + 1 = 0$

Let  $f(x) = x \tan x + 1$

Take  $a = 2.5$ ,  $b = 3$

$$f(2.5) = 2.5 \tan(2.5) + 1 \approx -0.870$$

$$f(3) = 3 \tan(3) + 1 \approx 0.572$$

Since  $f(2.5)f(3) < 0$ , a root lies between 2.5 and 3.

**False Position Formula:**  $x = [a f(b) - b f(a)] / [f(b) - f(a)]$

$$\text{Iteration 1: } x_1 \approx [2.5(0.572) - 3(-0.870)] / [0.572 - (-0.870)] \approx 2.8013$$

| Iteration | a      | b      | x      |
|-----------|--------|--------|--------|
| 1         | 2.5000 | 3.0000 | 2.8013 |
| 2         | 2.5000 | 2.8013 | 2.7985 |
| 3         | 2.5000 | 2.7985 | 2.7984 |
| 4         | 2.5000 | 2.7984 | 2.7984 |

**Therefore,**  $x = 2.7984$  (correct to four decimal places).

**Final Answers**

| Question | Root   |
|----------|--------|
| (i)      | 2.7065 |
| (ii)     | 1.3688 |
| (iii)    | 2.1149 |

|        |        |
|--------|--------|
| (iv)   | 1.4036 |
| (v)    | 0.8526 |
| (vi)   | 0.5885 |
| (vii)  | 0.7391 |
| (viii) | 2.7984 |

Note: Trigonometric functions are evaluated in radians for (vi), (vii), and (viii).